NeuroBytes: EEG-to-video retrieval

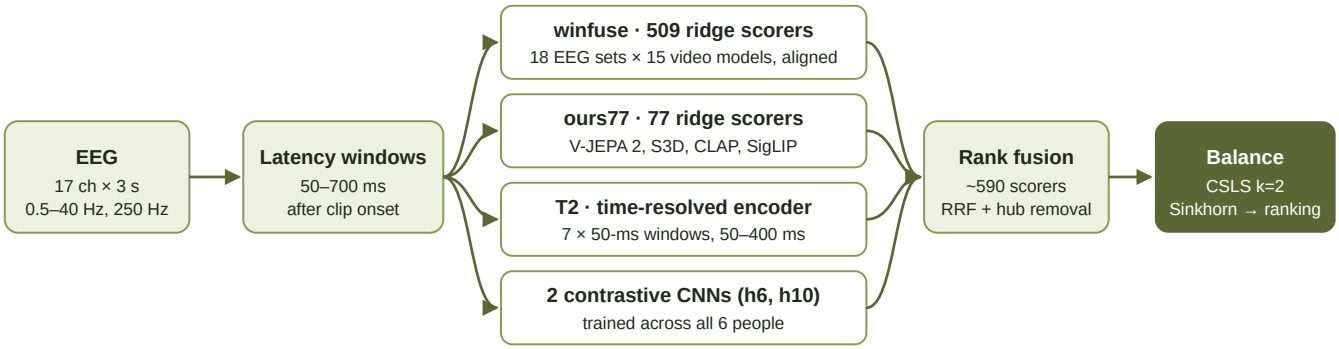
Team BierBytes: Sushruth Shivaraju, Ayush Shetty · SchubKI National AI Hackathon 2026, EEG mind-reading challenge · **github.com/Sushruths04/neurobytes**

Task. From a 3-second, 17-channel EEG recording over the visual cortex, rank the 102 candidate test videos so that the watched video comes first.

36.03%	**67.16%**	**5.99**
top-1 (chance 0.98%)	top-5 (chance 4.90%)	mean rank of 102 (chance 51.5)

Official development test, sub-01..04, 408 recordings. Final system: C13. Predictions: `submissions/C13/`.

1. Approach



- **Preprocessing:** band-pass 0.5–40 Hz (Butterworth, zero-phase), 50 Hz notch, downsample to 250 Hz, per-channel z-score from each person's training recordings.
- **Video features:** frozen pretrained models embed the candidate videos (AlexNet, ResNet-50, CLIP, DINOv2, SigLIP, VideoMAE, V-JEPA 2, S3D, CLAP, Whisper and others; exact versions in `MODELS.md`).
- **Scorers:** ridge regressions map windowed EEG to video features (PCA 150/256, penalty by 5-fold CV). For each person, other people's EEG is aligned into that person's channel space using the 695 videos everyone watched, then pooled.
- **Networks:** per-person 17→32 spatial layer, 3-layer temporal CNN, one head per video embedding; symmetric InfoNCE, AdamW, 40 epochs, 5 seeds; 6 and 10 heads.
- **Fusion and balancing:** scores are fused by reciprocal rank with hub correction. CSLS removes "hub" videos, and Sinkhorn makes each video win about once across a person's 102 recordings.

```
score = 0.85 × C10 blend + 0.075 × network h6 + 0.075 × network h10  →  CSLS (k=2)  →  Sinkhorn (3 iter., τ=0.5)
C10 blend = 0.85 × (509/586 × winfuse + 77/586 × ours77) + 0.15 × T2
```

All settings were chosen by cross-validation on held-out training videos. Python 3.11, NumPy, SciPy, scikit-learn, PyTorch 2.4. Weights: release `v1.0`. Docker: `ghcr.io/sushruths04/neurobytes`.

2. Results

How the score was built	Top-1	Top-5
Ridge baseline (one embedding)	11.0	29.7
winfuse: 509 latency-window scorers	23.0	52.5
+ 77 video-model scorers	26.5	57.4
+ hub correction (CSLS)	30.9	62.3
+ alignment, T2, Sinkhorn	32.6	66.7
+ two contrastive networks (C13)	**36.0**	**67.2**

Participant	Top-1	Top-5
sub-01	49.0	73.5
sub-02	32.4	68.6
sub-03	26.5	57.8
sub-04	36.3	68.6

Robustness	Top-1	Top-5
Without Sinkhorn	32.35	66.67
Five new network seeds	34.31	67.65
Same-seed rebuild		bit-identical

Where the true video lands: rank 1 in 36.0% of recordings, rank 2 in 10.3%, ranks 3–5 in 20.8%, ranks 6–10 in 16.9%, beyond rank 10 in 15.9%.

3. Failure cases

- **Near misses dominate.** About half of the misses still have the correct video in the top 5. The model usually finds the right neighbourhood, and the noise in a single recording decides the last step.
- **Misses are noise, not hard videos.** Per-video difficulty barely correlates between people (r = 0.16), so no fixed set of videos fails for everyone.
- **One participant is harder.** sub-03 reaches 26.5% / 57.8%, against 32–49% / 69–74% for the others, because of lower signal quality. Four targeted fixes all lowered accuracy, so none was kept.
- **Late signal is unusable.** Decoding from any window after about 800 ms is at chance, even when locked to the frame on screen.
- **Approaches that did not help:** hard one-to-one assignment (Hungarian) instead of Sinkhorn, banded ridge, pooling other people's EEG without alignment (−7.05 points), larger seed and window ensembles, per-person network fine-tuning, and caption features.

4. Lessons learned

- **When to look matters more than model size.** Restricting the EEG to 50–400 ms was the largest single gain (winfuse top-5 39.95% → 47.30%). Mid-level CNN layers (ResNet-50 layer 3, AlexNet fc6) match this window better than the outputs of larger models.
- **Use the structure of the task.** Each video is watched once, so balancing the score matrix with Sinkhorn added 6.13 points of top-5 without touching the EEG.
- **Correct for hubs and fuse ranks.** Rank fusion lets hundreds of scorers be combined without calibration. CSLS added 4.4 points of top-1.
- **Learn across people only after aligning them.** Every head mixes the brain's signals differently. Aligned pooling helps every participant; raw pooling hurts.
- **Many simple models beat one complex one.** About 590 closed-form ridge scorers and two small CNNs, chosen on validation, gave a stable result that rebuilds bit-identically.

Reproduce: `python combo/c13.py --dev` (seconds, shipped component scores) · `python scripts/reproduce_c13.py --data <challenge folder>` (networks recomputed from raw EEG) · full rebuild: `PIPELINE.md`. Web UI: `ghcr.io/sushruths04/neurobytes-web`.